

Docker practical questions:

1. Run a Docker container named "DB-app" based on the "mongodb" image, and expose port 80 on the host to port 8082 on the container ? Answer:- `docker run -d --name DB-app -p 80:8082 mongo`
2. You are a devops engineer in a startup company. The development team asks you to quickly deploy a simple web page that displays the message?
 1. use official httpd Docker image
 2. Run the container so it is accessible on port 8080 of host machine
 3. create or update an HTML file inside the container to display the required message
 4. verify the output using a command-line tool
1. How would you use the docker run command with -it, -e, -v, and --name to:
 1. set an environment variable APP_ENV=production.
 2. Bind a local directory /app/data to /data inside the container.
 3. Name the container my_app.

Solution

```
docker pull ubuntu
```

```
mkdir C:\app data
```

```
docker run -it --name my_app -e APP_ENV=production -v C:\app\data:\data ubuntu /bin/bash
```

```
echo $APP_ENV
```

Task:_____

1. Inside this same running container, you are required to

.create a new directory name/project

.inside project, create a file called report.txt

.Add the following text inside the file

This is our container interaction with our host machine.

.verify file content

.copy report.txt from the container to your desktop

.create another file named notes.txt in the same project directory

.Append the text

.verify that both files exist inside project

questions:

1. run a docker container named "DB-app" based on the "mongodb" image, and expose the port 80 on the host to port 8082 on the container.

1. you are a devops engineer in a startup company. the development team asks you to quickly deploy a simple web page that displays the message:

use the official httpd Docker image.

run the container so it is accessible on port 8080 of the host machine.

create or update an HTML file inside the container to display the required message.

verify the output using a command line tool.

properly stop and remove the container after testing.

1. how would you use the docker run command with -it,-e,-v,and --name to:

- set an environment variable APP_ENV=production

-bind a local directory/app/data to/data inside the container.

-name the container my_app

Note: find ~ -name "test.txt"

4) inside the same running container, you are required to:

1. create a new directory named /project
2. inside /project, create a file named report.txt
3. add the following text inside the file: this is our container with our host machine.
4. verify the file content
5. copy report.txt from the container to your desktop
6. create another file named notes.txt in the same /project directory
7. append the text
8. verify that both files exist inside /project
1. you need to start a new container using the nginx image while setting an environment variable ENV_MODE=production. write the docker run command to achieve this.
2. you have a running container named my_app, but it's not behaving as expected. you want to check its logs to debug any errors. write the command to view its logs and follow new log entries in real-time.
3. a container named web_server was stopped. write the command to: a) start the container again. b) stop the container when needed.

docker volume create <volume_name>

docker volume ls

docker volume inspect mydata

docker run -d -v <volume_name>:<path where to store> --name <container_name> container

docker inspect mycontainer

docker rm -f mydata

docker exec -it mycontainer ls /app/data

docker run -dit --name mycontainer -v mydata:/app/data ubuntu

docker volume prune

8. a. create a docker volume named projectdata
- b. verify that the volume has been created successfully
- c. inspect the volume and identify: the mount point(where docker stores it on the host system)
- d. run an ubuntu container in detached mode: name the container project_container
- e. attach the volume projectdata inside the container at /app/data
- f. inside the running container